Polynomial bit rare deadline probability for exponential precedence networks

Proof candidate draft
12 October 2026

This document gives a candidate randomized polynomial-bit scheme for additive approximation of the logarithm of a deadline-overrun probability in an explicitly represented directed acyclic precedence network with independent exponential activity durations. Its only external algorithmic theorem dependency is the published #NFA FPRAS [1]. The construction is independent of the earlier saturated-transport candidate.

The candidate has received a separate mathematical check that reported no fatal gap after the implementation guards were incorporated. It is not formally verified or externally certified. No novelty, validated software, or practical performance claim is made.

Theorem and scope

Theorem (anchored decreasing-density/NFA log-tail scheme). Consider a finite DAG with independent exponential random durations on its activities, strictly positive rational rates supplied exactly in binary, and a rational deadline T supplied in binary. Activities may be edges, or vertices represented by the standard vertex-splitting construction. Deterministic zero-duration connectors are permitted. Let L be the longest source-target duration, and U=Pr(L>=T). For rational epsilon>0 and 0<delta<1, there is a randomized algorithm whose running time and bit complexity are polynomial in the DAG size, total input bit length, 1/epsilon, and log(1/delta), and which, whenever U>0, outputs a rational ell satisfying

```
Pr[ |ell-log U| <= epsilon ] >= 1-delta.
```

If U=0 it returns the exact symbol -infinity. If T<=0 and a source-target path exists, it returns ell=0. A missing source-target path is treated as U=0 by convention. For T>0, deterministic zero-only paths do not contribute; if no path contains a positive-duration activity, U=0. Otherwise the existence of a relevant exponential activity guarantees U>0.

One conservative parameter choice is epsilon0=min(epsilon,1/10), tau=epsilon0/(10^6 K), finite-precision failure allowance zeta=epsilon0/(10^4 K), and N=ceil(5000 K^2 epsilon0^(-2) log(8/delta)) union samples. Budget each component arithmetic and accepted-word generation error at tau, splitting arithmetic tolerances over its polynomially many operations. The bounds above then give component relative-count and ideal conditional-law errors below 20tau, while Hoeffding gives union-mean absolute error at most epsilon0/(100K) with failure at most delta/4. Allocate total failure delta/4 to the component normalizers and compute the final rational logarithms to additive epsilon0/10. These deliberately loose constants leave the total log error below epsilon. Count-oracle confidence is amplified and allocated over the polynomial number of normalization calls; within each generated sample, prefix-count errors are budgeted to total variation tau as in Section 10. This gives polynomial dependence on log(1/delta), rather than inverse delta.

The theorem is for independent exponential activities that occur at most once on a DAG path. It does not claim an extension to correlated durations, shared physical activities repeated across multiple path positions, positive deterministic offsets, or arbitrary finite-atomic two-budget problems. Vertex splitting preserves independence and introduces only zero connectors. When zero connectors are present, the skip/select automaton scans every edge topologically but updates the random-coordinate and resource counters only on exponential-activity layers; zero transitions are epsilon-eliminated or merged into the intervening deterministic relation. Thus there are still polynomially many states and each independent activity is read exactly once.

Construction and proof

1 Exact polynomial size union of anchored components

Let relevant activities be independent X_j ~ Exp(lambda_j), all rates and T>0 rational in binary. Set A={some s-t path has duration at least T}. Order activities by (rate,index). For each i retain only activities with rank at least rank(i), and

require the path to contain i. Empty classes are deleted. These path classes partition all original paths, without enumerating them. The graph for a class is polynomial size (prefix to i, i, suffix from i). The rate ranking only filters admissible activities. The skip/select automaton scans the surviving activities in a fixed topological path-consistent order, not in rate order.

Let V_i be the union of activities on paths of class i. Put

```
A_i = {a class-i path overruns T} intersect {X_j<T for every j in V_i except i}.
  B_j = {X_j>=T}, for each relevant activity j.
```

Then exactly A = (union_i A_i) union (union_j B_j). Every component is contained in A. Conversely, if some coordinate reaches T a B_j holds; if none does, any overrun path supplies an A_i. There are at most 2n components. Component membership is polynomial-time longest-path computation plus coordinate comparisons.

For A_i write lambda=lambda_i, m=|V_i|-1, S=sum_(j != i) X_j, and L=max class-path duration excluding X_i. Thus 0<=L<=S. For j in V_i except i, put

```
delta_j=lambda_j-lambda >=0,
  I_j=integral_0^T exp(-delta_j x) dx,
  C_j=lambda_j I_j,
  q_j(x)=exp(-delta_j x)/I_j on [0,T].
```

For delta_j=0 this is the uniform distribution. All q_j are decreasing densities. Integrating X_i gives the exact identity

```
Pr(A_i)=exp(-lambda T) product_j C_j * p_i,
  p_i=E_q exp[-lambda R(X)],
  R(X)=S-min(L,T)=max(S-L,S-T)>=0.
```

With independent E~Exp(lambda),

```
p_i=Pr_q,E[ min_P sum_(j not in P) X_j <= E AND S<=T+E ].
```

The min ranges over class-i paths with activity i suppressed. The omitted-cost condition has the standard ordered skip/select layered automaton; the total-cost resource is path-independent.

For m=0, Pr(A_i)=exp(-lambda T) directly. Below m>=1.

2 Polynomial bit lower bound and density bound

Set M=max(1,m lambda T) and rho=1/(4 M^m). Then p_i>=rho. Indeed take h=min(T,1/(m lambda)); every decreasing density on [0,T] has Pr(X_j<=h)>=h/T. On this event S<=1/lambda, and E>=1/lambda suffices. Therefore p_i>=exp(-1) M^(-m)>rho.

The rational rho has polynomial bit length. Also

```
sup_x q_j(x)=delta_j/(1-exp(-delta_j T)) <= delta_j+1/T,
```

with the delta=0 value interpreted as 1/T. This follows from 1-exp(-z)>=z/(1+z).

For stable rate arithmetic write C_j=lambda_j*T*phi(delta_j*T), where phi(z)=(1-exp(-z))/z for z>0 and phi(0)=1. Use the Taylor expansion of phi for small z; for large z compute log C_j=log(lambda_j/delta_j)+log(1-exp(-z)) and discard exp(-z) only at the required certified absolute accuracy. Thus C_j and log C_j are computable to the required relative/additive accuracy with polynomial bit work, including equal and near-equal rates and huge binary products. Their logarithms have polynomial magnitude in the input bit size. The huge numerical exponent remains the explicit binary rational -lambda T.

3 Simultaneous scaling lemma

For e,b>=0 define the downward event

```
D(e,b)={there exists a class path with omitted sum<=e, and S<=b}.
```

For every 0<a<=1,

```
Pr_q[D(ae,ab)] >= a^m Pr_q[D(e,b)].
```

Proof: x -> ax maps D(e,b) inside D(ae,ab), stays in [0,T]^m, has Jacobian a^m, and q_j(ax_j)>=q_j(x_j). This is the crucial relative slack bound. It is not valid for arbitrary finite-atomic marginals, and does not assert a general two-budget finite-atomic theorem.

4 Comparator discretization

Choose rational 0<tau<=1/100. Let h_E=tau/lambda. Take L0=1+ceil(log_2(1/(tau rho))), computable from rational bit lengths (or any explicitly polynomial-size upper bound of this order), and K=ceil(L0/tau); H=K h_E. Map E<H to e_k=k h_E for k=0,...,K-1 and reject E>=H. Thus

```
Pr(floor(E/h_E)h_E>=r, E<H)
  >= exp(-tau) exp(-lambda r)-exp(-lambda H)
```

for every r>=0, and is at most exp(-lambda r). Consequently the resulting remainder p_grid lies in

```
exp(-tau) p_i-tau rho <= p_grid <= p_i.
```

K is polynomial in input bits and 1/tau. The E-cell masses are exp(-k tau)-exp(-(k+1)tau), plus a rejected tail symbol.

5 Geometric atomization and cumulative resource rounding

Let d_max=max_j(delta_j+1/T) and ell=tau rho/(m d_max). Then Pr_q[some X_j<ell]<=tau rho. Take rational g=1+gamma with gamma = tau/[10m(m+1)] exactly. The explicit equality ensures that the number of geometric grid points is polynomial; an unspecified arbitrarily smaller gamma would not suffice. Geometrically round every X_j upward to an atom in {ell g^r:r>=0}, stopping at the first grid point >=T; values below ell are clamped to ell. Off the bad event X_j<ell, each atom is <=g X_j.

Use the same geometric grid plus zero for both accumulated resources. On a skip transition add the current atom to omitted cost; on a select transition add zero. On every transition add the atom to total cost. After each addition round UP to the next resource grid point. Zero remains zero, and adding zero to an existing grid point leaves it unchanged. Induction gives, for every selected path and every input outside the bad event,

```
exact omitted <= rounded omitted <= g^(m+1) exact omitted,
  exact total   <= rounded total   <= g^(m+1) exact total.
```

Let D0=g^(m+1). The rounded automaton for comparator e_k accepts only D(e_k,T+e_k), and contains D(e_k/D0, (T+e_k)/D0) outside the bad event. By the smoothing lemma its acceptance probability Q_round,k satisfies

```
D0^(-m) Pr_q[D(e_k,T+e_k)]-tau rho
  <= Q_round,k <= Pr_q[D(e_k,T+e_k)].
```

Average over the comparator cells. Since m(m+1)log(g)<=tau/10, total relative geometric-rounding loss is O(tau). Together with comparator discretization,

```
(1-O(tau)) p_i <= p_round <= p_i.
```

Keep resource grid values through the first point at least 2mT (which exceeds g^(m+1)mT for the chosen gamma), plus zero. There are only O(log(2mT/ell)/log(g)) positive resource grid values. Each has polynomial rational bit length. The automaton state stores (original selected endpoint, rounded omitted resource, rounded total resource, comparator index). The number of states and transitions is polynomial. Overflow is an absorbing reject state. A zero-cost forced transition for the anchored activity can be removed by ordinary epsilon-elimination or a fixed bridge between prefix and suffix; it carries no random coordinate.

This is an explicit approximate nondeterministic read-once branching program. It does not count witnesses; different paths accepting the same input are accepted only once by #NFA.

6 Rational laws and reduction to ordinary NFA

The atom masses are differences of truncated-exponential CDF values at rational endpoints. Approximate each coordinate law by dyadic masses with total variation <=tau rho/[10(m+1)]. This needs only polynomial absolute precision because log(1/rho), alphabet size, and all endpoint bit lengths are polynomial. Large exponential arguments

can be certified negligible at that absolute precision; cancellation in delta near zero is handled by stable normalization/integral evaluation. Zero dyadic masses can be removed.

Use a common denominator 2^b, with b polynomial. A uniform b-bit block is decoded to atom j according to a disjoint integer interval of length equal to its numerator. Each interval has a polynomial-size binary comparator automaton. Replace each atom-labelled transition by such a b-bit interval recognizer. The accepted bit strings encode the product atom law exactly. Multiple nondeterministic witness paths do not multiply the count of a bit string.

The resulting ordinary NFA and unary input length are polynomial. Its accepted-word fraction equals the rounded probability under the dyadic law. Total variation changes the event probability by at most tau rho/10, hence only O(tau) relatively. Invoke the #NFA FPRAS at relative accuracy tau. Taking logs and restoring -lambda T+sum log C_j approximates log Pr(A_i) to O(tau).

The external counting theorem [1] counts accepted words, not accepting runs, and also supplies polynomial-time generation. An approximate conditional generator can alternatively be built from prefix-count self-reduction.

7 Conditional samples and the polynomial union correction

For a generated accepted atom word, jitter each X_j independently according to the original q_j law restricted to its atom interval, and E according to its comparator cell. For the exact atom masses this produces q times the independent E law conditioned on the rounded acceptance event. That acceptance event is a subset of {E>=R}; its mass differs by O(tau) relatively, so conditioning changes the desired tilted joint law by O(tau) in total variation. Dyadic-law and NFA-generator errors can also be budgeted as O(tau), because the normalizer is >=rho/2.

Project to the non-anchor X_j, then sample

```
X_i=(T-L)_+ + Z, Z~Exp(lambda),
```

and sample every activity outside V_i independently from its original exponential law. This is within O(tau) total variation of the original environment conditional on A_i. The true ideal conditional law follows directly from the likelihood identity in Section 1. B_j conditionals are elementary memorylessness.

Now use the ordinary union-of-at-most-2n-events estimator: select a component with probability proportional to its probability, sample its conditional environment, evaluate the number M(X) of true components containing X, and average 1/M(X). The ideal mean is U/W and lies in [1/(2n),1]. Choosing all component count and conditional-sampling errors tau=O(epsilon/n), with appropriate constants and polynomial sample count, gives relative O(epsilon) error in U and hence additive O(epsilon) error in log U. Component weights are normalized in log scale without ever writing exp(-lambda T) as a huge denominator. Optional omission of negligible components must use the retained-component membership count throughout. Require total discarded component weight at most epsilon*W/(100K), which ensures lost union mass at most epsilon*U/100 because W<=K*U. It suffices to drop only components whose true log gap from the largest component is at least log(100K^2/epsilon); when gaps are computed approximately, add an explicit guard at least twice the known log-estimation error. Alternatively keep every component and implement log-scale categorical sampling to controlled total-variation accuracy.

Finite precision sampling

A discrete atom sample is not itself a continuous conditional sample. The jitter must use lazy continuous random bits or an explicitly bounded finite-precision coupling. Membership checks for the true A_i/B_j can use interval arithmetic and be refined until unambiguous, except on null boundaries. A worst-case polynomial-bit truncation uses the following anti-concentration argument; the product tilted densities have polynomial-bit logarithmic sup bounds and conditioning normalizer >=rho/2, and the shifted anchor adds an independent exponential. Section 9 below supplies a polynomial precision bound from these density and conditioning estimates; continuous sampling and exact-real comparisons are not assumed as oracles.

8 Scope of the reduction

The two-resource finite-atomic transport is not assumed. The geometric rounding is justified before atomic discretization by decreasing continuous densities and simultaneous scaling of all coordinates. The full heterogeneous original tail does not have that slack property; integrating a minimum-rate anchor converts it to a downward event

under decreasing product densities. Path multiplicity is handled by the established accepted-word #NFA algorithm inside each component, and the remaining component multiplicity is at most 2n.

9 Finite precision membership and sampler implementation

Here is an explicit anti-concentration bound for the ideal jittered sampler from Section 7. Its conditioning is the rounded acceptance event under the exact q/E product law; its probability is at least rho/2 for sufficiently small tau. Before that conditioning, non-anchor class coordinates are independent q_j variables on [0,T], and coordinates outside V_i are independent original exponentials. The anchor is then X_i=(T-L_i)_+ +Z with independent Z~Exp(lambda_i).

Let D=max(1,max_j lambda_j,max_(j in V_i except i)(delta_j+1/T)). For any fixed nonempty original path P and any u>0:

- If P contains i, conditional on all other base variables its duration is Z plus a fixed shift. Therefore Pr(|duration(P)-T|<=u)<=2 lambda_i u.
- If P does not contain i, choose any j in P. Under the unconditioned independent base law its sum has density at most D, by conditioning on the other coordinates. Imposing the rounded-acceptance conditioning raises this strip probability by at most 2/rho. Hence Pr(|duration(P)-T|<=u)<=4Du/rho.

The same bounds apply to strips |X_j-T|<=u. A DAG with n activity edges has at most 2^n distinct path edge sets. Across at most n anchored component tests, a coarse union bound therefore makes every true component-membership test unambiguous, except with probability zeta, when

```
u = zeta rho / [100 n 2^n D].
```

This rational has polynomial bit length. Empty paths have length zero and cannot meet T>0, so they create no boundary issue. For a B_j conditional sampler, memorylessness gives the analogous bounds directly, without the rho denominator.

Compute each non-anchor base coordinate and Z to an enclosing interval of width at most u/(10n^2). Longest-path and positive-part computations are 1-Lipschitz in the sum norm, so the anchor interval and then every path-sum interval have width less than u. If any component-membership test remains ambiguous, return an arbitrary indicator or sample and charge the entire event to failure zeta. This is a controlled finite-precision coupling; no assertion of TV proximity between a discrete vector and a continuous law is made.

All required coordinate magnitudes have polynomial bit length with probability 1-zeta: class coordinates are bounded by T; outside exponentials and Z can be capped for computation at J/lambda_j, where integer J>=log(10n/zeta), at total exceptional probability O(zeta). The anchor is at most T+Z.

Conditional jitter of a truncated exponential on a rational interval [a,b] is explicit. For delta>0 use

```
a - (1/delta) log(1-U(1-exp(-delta(b-a)))), U uniform(0,1),
```

and for delta=0 use a+(b-a)U. If delta(b-a) exceeds a polynomial logarithmic precision cutoff, replace the interval law by a+Exp(delta), charging its exponentially small truncation tail. Otherwise certified exp/log or monotone CDF bisection computes the displayed quantile. Excluding U within kappa of an endpoint costs at most 2kappa and bounds the required inverse-CDF derivative; all logs of the input scales, interval widths, target width, and kappa are polynomial. Choosing kappa=zeta/poly(n) gives a worst-case polynomial-bit sampler whose generated intervals couple to the ideal continuous variables within the requested widths except with probability O(zeta).

Dyadic atom-law error and approximate NFA word generation are first coupled at the atom-word level. On agreement the same conditional jitter kernel is used, so their O(tau) TV error remains O(tau) before finite-precision comparisons. The above anti-concentration analysis is needed only for the ideal exact-law word sampler.

10 Explicit robust union estimator error accounting

Let K<=2n be the number of nonempty components, p_r their true probabilities, W=sum p_r, and pi_r=p_r/W. Suppose every estimated p_r has relative error at most a, and every conditional sampler is within TV b of the true conditional law. The estimated selection probabilities differ from pi in TV at most 2a (for a<=1/4). The ideal variable Z=1/M(X), where M counts the true containing components, lies in [1/K,1] and has mean U/W>=1/K. Approximate selection and sampling change its mean by at most 2a+b, and membership-computation failures at probability zeta add at most zeta.

If finite-precision membership produces M=0, define the observed estimator to be 0; this bounded fallback is charged to the same anomaly probability.

Take a,b,zeta<=epsilon/(100K), estimate this bounded mean with O(K^2 epsilon^-2 log(1/delta)) samples, and combine with the relative approximation to W. This gives relative O(epsilon) approximation to U with the requested constant or amplified success probability, hence additive O(epsilon) accuracy for log U. All calls use only inverse-polynomial accuracy in n and epsilon, so the construction is fully polynomial. Numerical evaluation returns log W+log(mean), not an explicit huge-denominator probability.

Approximate accepted-word generation needs no additional theorem beyond #NFA approximate counting: for a binary word of length ell_word, self-reduce one bit at a time using counts for each restricted prefix, with relative accuracy O(b/ell_word) and failure O(b/ell_word) per call. Zero prefixes are detected by NFA reachability. The accumulated TV error is O(b), and all restricted automata stay polynomial size.

11 Edge cases and deterministic checks

Remove activities on no source-target path. If no nonempty path remains and T>0, U=0 and log U=-infinity; if T<=0 and a path exists, U=1. Empty paths cannot overrun a positive deadline. Empty anchored path classes are skipped, and classes with no non-anchor coordinates are single exponential tails.

Reported deterministic checks exhaustively checked the exact anchored-union identity on 15,625 assignments for a six-edge DAG with tied and unequal rates, and the simultaneous resource-rounding inequalities on 10,000 four-coordinate/select-pattern combinations. All reported assertions passed. These are algebra/implementation spot checks, not a replacement for the proof and not Monte Carlo evidence.

The resource grid extends through its first point at least 2mT, safely above D0*mT, rather than stopping at m*g*T. The cap is essential even though the difference is only a small multiplicative factor.

Implementation conventions

Select an edge only when it continues the currently selected path. Track the selected endpoint and the anchor-required condition or an equivalent prefix/suffix phase. Enforce the designated source and target. Compute the relevant activities by reachability; do not enumerate paths or determinize subsets.

All displayed logarithms are natural logarithms. The symbol K is local: in Section 4 it counts comparator cells; in the theorem and Sections 7 and 10 it counts nonempty union components. The final output is a binary rational approximation to log U. An extremely small U is never represented by an exponentially long probability denominator.

Use gamma=tau/[10m(m+1)] exactly, cap tau by a fixed small constant, and retain the resource grid through its first point at least 2mT. Split arithmetic tolerances across the polynomial number of operations. Handle m=0, no feasible path, zero-count cases and confidence failures separately. Exceptional membership outcomes with M=0 receive bounded estimator value zero. A finite rational sample is not claimed to be close in total variation to a continuous distribution; finite enclosures are coupled to ideal samples, and only the needed membership decisions are controlled.

If negligible components are discarded, apply the guarded K-dependent cutoff and use retained-component membership throughout. Retain deterministic zero-duration connectors in vertex-split graphs without assigning them rate ranks or random coordinates. Cycles, correlated durations, repeated uses of one random activity, positive deterministic offsets and arbitrary implicitly specified path families are outside the stated theorem.

Reference

[1] Marcelo Arenas, Luis Alberto Croquevielle, Rajesh Jayaram, and Cristian Riveros. #NFA Admits an FPRAS: Efficient Enumeration, Counting, and Uniform Generation for Logspace Classes. Journal of the ACM 68(6), article 48, 2021.

DOI: https://doi.org/10.1145/3477045
Full text: https://marceloarenas.cl/publications/jacm21.pdf
Preprint: https://arxiv.org/abs/1906.09226